

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	P Tanvitha
Roll Number	2520030385
Branch	CSE
Course Outcome	CO-6

Case Study Topic: Bangalore-to-Mumbai Logistics — 0/1 Knapsack for Cargo Loading

Work Done Summary:

Applied Greedy strategy (sort by value/weight ratio and pick until capacity full) to the 8 consignments.

Demonstrated that greedy is not optimal for 0/1 Knapsack by showing a different valid subset with higher value.

Computed the optimal value via 0/1 Knapsack DP for confirmation.

Implementation Details:

- Eight consignments with given weight (tons) and value (₹-thousand).
- Truck capacity $W = 24$ tons.
- Greedy: sort by value/weight ratio descending and pick while capacity allows.
- Brute-force DP (0/1 Knapsack) used to compute optimal value and optimal subset.

Result: Screenshots / Output

```
PS C:\DSA-2> javac KnapsackLogistics.java
PS C:\DSA-2> java KnapsackLogistics

BANGALORE-TO-MUMBAI LOGISTICS - 0/1 KNAPSACK (CAPACITY = 24 TONS)
-----
Items (Consignments):
Item  Weight(tons)  Value(₹k)  Value/Weight
-----
A      5             40         8.00
B      8             50         6.25
C      3             20         6.67
D     10             70         7.00
E      4             30         7.50
F      6             35         5.83
G      7             45         6.43
H      2             15         7.50

GREEDY ATTEMPT (Sort by value/weight ratio descending)
-----
Order of consideration (by v/w ratio):
A(8.00), H(7.50), E(7.50), D(7.00), C(6.67), G(6.43), B(6.25), F(5.83)
Items packed by Greedy:
A, H, E, D
Total weight = 21 tons / 24 tons
Total value  = ₹155k

A BETTER VALID SUBSET (Not by greedy)
-----
Items: D, B, C, E
Total weight = 10 + 8 + 3 + 4 = 21 tons / 24 tons
Total value  = 70 + 50 + 20 + 30 = ₹170k

OPTIMAL SOLUTION (0/1 Knapsack DP)
-----
Optimal value = ₹170k
Optimal items = D, B, C, E
Total weight = 21 tons / 24 tons
```

Greedy picks {A, H, E, D} with value ₹155k at weight 21 tons.

But subset {D, B, C, E} gives higher value ₹170k within the 24-ton limit.

Thus, greedy is not optimal for 0/1 Knapsack; DP confirms optimal value = ₹170k.

GitHub Link: <https://github.com/pondurutanvitha-stack/DSA-2>

Student Signature	Faculty Signature